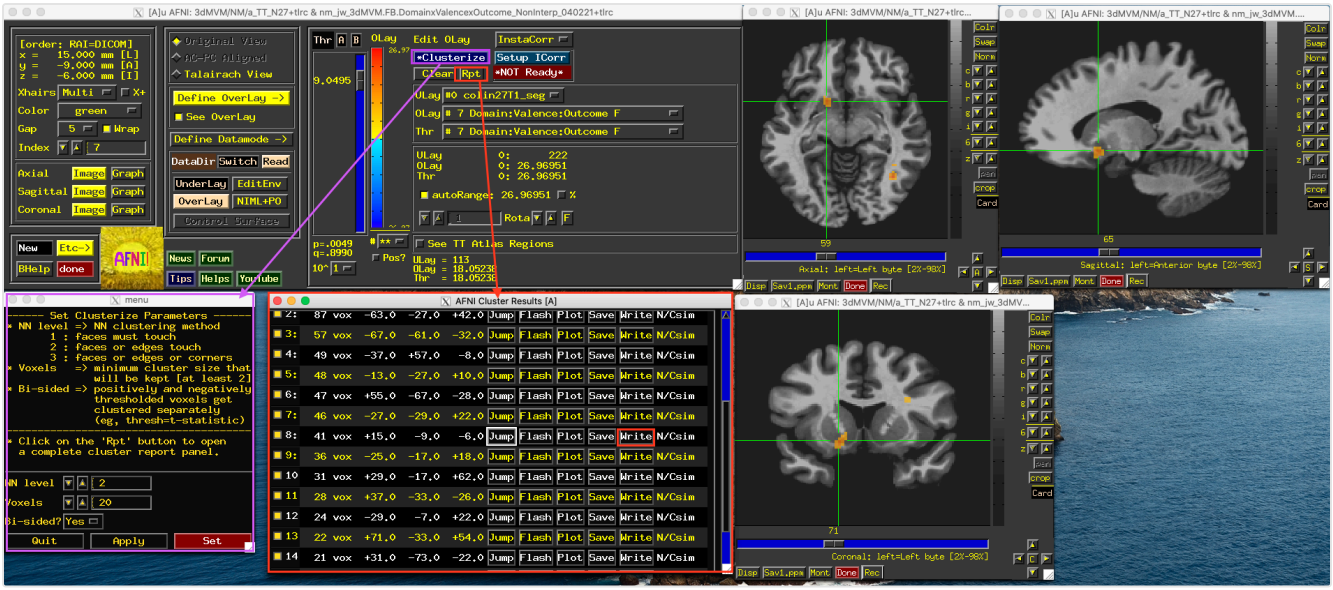
Extracting ROIs

This page outlines how to generate masks (briefly) from an atlas, then covers the steps for extracting ROIs and converting output to .csv format

Masks

Activation-Based

Write a mask directly from an activation cluster in your 3dMVM output's 'clusterize' report



Anatomical

Identify the atlas you'd like to extract a region from. AFNI has several pre-loaded, you can also download an atlas from here for anatomical extraction.

> *If downloading from the AFNI repository...*
> *1. Download file into folder where ROIs will be extracted.*
> *1b. Extract the contents of folder if necessary*
> *2. Each atlas comes with a text file noting the label for each region (e.g. 1 Precentral_L)*
> *Note the number labels for each ROI*

> *3. Use 3dcalc (AFNI command) in command line. This code will isolate the ROI and "zero" out any thing in the atlas which is not the specified ROI.*
> *4. If you would like to make a bilateral mask for each ROI then use 3dcalc in the command line.*

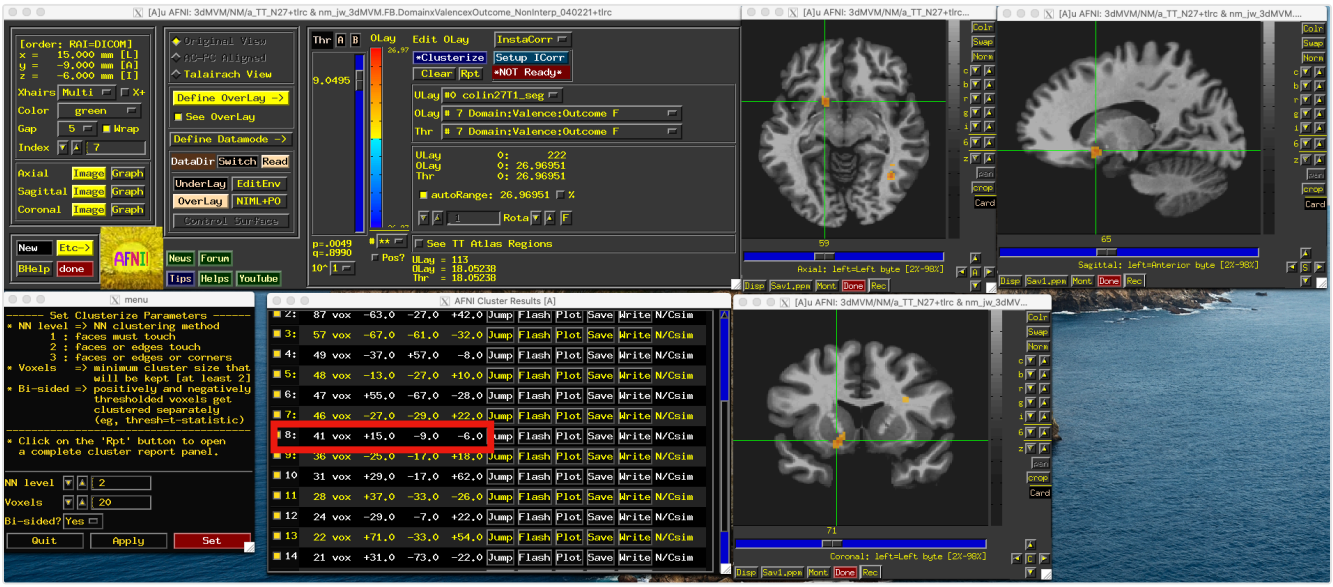
```
# Example text for command line:
cd ROI/Directory
wget url-to-atlas
tar -xzf atlas_file.tgz
# ROI= 155 L_amygdala 156 R_amygdala
3dcalc -a Atlas_file.nii -expr 'equals(a,155)' -prefix amygdala_left_atlas.nii
3dcalc -a Atlas_file.nii -expr 'equals(a,156)' -prefix amygdala_right_atlas.nii
# Add hemispheres for bilateral mask
3dcalc -a amygdala_left_atlas.nii -b amygdala_right_atlas.nii -expr '(a+b)' -
prefix amygdala_lr_atlas.nii
```

If the atlas you'd like to extract from is loaded in AFNI, you can use the whereami command to extract these regions as well

```
# Example text for command line:
whereami -mask_atlas_region TT_Daemon::9 -prefix mfg
# This command would extract BILATERAL middle frontal gyrus, from Broadmann area 9
(and it's hemispheric correlate)
```

Spherical

Generate a spherical mask based on a voxel of peak activation of an ROI from your 3dMVM output's 'clusterize' report. The report's default displays the coordinates for the peak voxel automatically:



When you have the coordinates, use AFNI's 3dUndump command to generate your sherical mask:

```
# Example text for command line:
echo "15 -9 -6" | 3dUndump -prefix VS_8mm_l -srad 8 -orient RAI -master
nm_jw_3dMVM.FB.DomainxValencexOutcome_NonInterp_040221+tlrc. -xyz -
```

Manipulating Masks

Once you've pulled masks, there are many relevant ways to edit them with 3dcalc or 3dmask_tool

- Add separate masks together (e.g. creating bilateral mask, creating construct network mask)

```
# Add hemispheres for bilateral mask
3dcalc -a amygdala_left_atlas.nii -b amygdala_right_atlas.nii -expr '(a+b)' -
prefix amygdala_lr_atlas.nii
# Add anatomical masks to create a construct mask (e.g. Threat mask)
3dcalc -a amygdala_lr.nii -b insula_lr.nii -c dACC_lr.nii -expr '(a+b+c)' -prefix
threat_mask.nii
```

- Create subsections of a mask using axis thresholds (e.g. bisect a mask, make a bilateral mask hemisphere-specific)

```
# Bisect a mask on the Y-AXIS, everything anterior to y=9
3dcalc -a dACC_left_atlas.nii -expr 'a*step(9-y)' -prefix
dACC_left_bisected_atlas.nii
# Everything posterior to y=9
3dcalc -a dACC_left_atlas.nii -expr 'a*step(y-9)' -prefix
dACC_left_bisected_atlas.nii
# Split bilateral mask into hemispheres
3dcalc -a amygdala_lr_atlas.nii -expr 'a*step(0-x)' -prefix amydala_r_atlas.nii
3dcalc -a amygdala_lr_atlas.nii -expr 'a*step(x-0)' -prefix amydala_l_atlas.nii
```

- Trim or augment around the borders of a mask by a number of voxels (e.g. fine-tune mask boundaries if it reaches into white matter or ventricles)
  - Sampling (voxel definition, e.g. 1x1x1) matters a lot here! If your mask is sampled at 0.5x0.5x0.5 and you erode by 3 voxels, this will be equivalent to eroding by 1.5 voxels when mask is resampled up to 1x1x1
  - Most atlases are sampled at 1x1x1, double-check with 3dinfo mask.nii

```
# Trim a mask by 1 voxel
3dmask_tool -inputs amygdala_lr_atlas.nii -prefix amygdala_lr_erode1_atlas.nii -
dilate_inputs -1
# Trim a mask by 1.5 voxels (1x1x1)
3dresample -master dxyz 0.5 0.5 0.5 -prefix amygdala_lr_050505_atlas.nii -input
amygdala_lr_atlas.nii
3dmask_tool -inputs amygdala_lr_050505_atlas.nii -prefix
amygdala_lr_050505_erode3_atlas.nii -dilate_inputs -3
3dresample -master stats.s(IndvlLvlAnlysSubj#)+tlrc -prefix
amygdala_lr_resampled_erode1_5_atlas.nii -input
amygdala_lr_050505_erode3_atlas.nii
# Augment a mask by 1 voxel
3dmask_tool -inputs amygdala_lr_atlas.nii -prefix amygdala_lr_dilate1_atlas.nii -
dilate_inputs 1
```

- 3dcalc Help Page
- 3dmask_tool Help Page

Extracting Data

June 4, 2021 [JW]: these scripts are going to be specific to the Social Doors College SBU data:

- Copy mask to `…/fMRI_ANALYSES/MASKS/ActivationBased/` (i.e., the MASKS/ subdir corresponding with your mask type
- Change directories to the individual level analyses `…/fMRI_ANALYSES/ IdvlLvlAnalyse s/` and create a `subjList.ROI._____` file that will be referenced in your Extraction script

```
s3836 s3845 s3846 s3847 s3848 s3849 s3851 s3852 s3855
s3864 s3865 s3871 s3877 s3880 s3882 s3883 s3886 s3887
s3889 s3890 s3891 s3892 s3893 s3895 s3896 s3910 s3912
s3914 s3920 s3967 s3992 s4017 s4018 s4019 s4020
~
~
```

- Open `…/fMRI_ANALYSES/SCRIPTS/ROI/nm_jw_RoiExtraction.sh` and make the following edits:
  - set collection

```
##########################################################################################
# GENERAL SETUP
##########################################################################################

# Specify interum date that data were collected through or sample sizes
set collection = 060221

# IMPORTANT NOTE: this script makes a list of everyone who has a flder in IndvlLvlAnalysis, and pulls data from thos
e people
# - TO ENSURE YOU PULL THE CORRECT PEOPLE FOR YOUR ANALYSIS, MODIFY YOUR LIST MANUALLY, OR TEMPORARILY MOVE DATA FRO
M INDVLVLANALYSES
```

  - Specify locations and information about your data
    - Important: for each new analysis, make sure you give it a new name (red box); this is the name of the output directory that will be created in the `…/fMRI_ANALYSES/MASKS/…/` directory

```
#####################################################
# SPECIFY YOUR LOCATIONS AND INFORMATION ABOUT YOUR DATA
#####################################################

# Specify the top-most directory associated with the study
set topdir = /data/projects/STUDIES/Social_Doors/TaskVersion_4Runs/Social_Doors_CollegeSBU/fMRI_ANALYSES

# Specify the where individual level analyses are found
set indvlvldata = $topdir/IndvlLvlAnalyses

# Specify the suffix for the individual level directory where your subject's stats files are located
#set subjresults = results.social.1mm
set subjresults = results.doors.1mm

## CHANGE ME:  Specify the analysis of interest (e.g., anticipation or feedback with which individual level data and
 date)
#set analysis = FB.SocialDoors.111319
#set analysis = Ant.SocialDoors.111319
#set analysis = FB.Doors.111319

set analysis = 3dMVM.FB.NonSocial_PositivexNMvstri_NonInterp_052321

#set atlasdir = /Volumes/SDN_Data/STUDIES/AFNI_ROI_Atlases/TT_Daemon_2.5x2.5x2.5
#set atlasdir = /Volumes/SDN_Data/AnatomicalROI_Masks/ROIs/TT_N27_EZ_ML
set atlasdir = /data/projects/STUDIES/Social_Doors/TaskVersion_4Runs/Social_Doors_CollegeSBU/fMRI_ANALYSES/MASKS/Ana
tomical/
```

- Specify information about your roi

```
#####################################################
# SPECIFY INFORMATION ABOUT YOUR ROI
#####################################################

# Specify type of ROI mask (Anatomical or ActivationBased)
set ROItype = Anatomical

## CHANGE ME: Specify the name of the anatomical ROI of interest

#TT N27 EZ ML
#set ROIs = ( Anterior_Cingulate_Cortex Caudate_Nucleus Fusiform_Gyrus Middle_Cingulate_Cortex Middle_Frontal_Gyrus
Middle_Orbital_Gyrus Middle_Temporal_Gyrus Pallidum Posterior_Cingulate_Cortex Superior_Temporal_Gyrus Temporal_Pole
 Thalamus )


#TT Daemon
set ROIs = ( VS_6mm )
#set ROIs = ( Amygdala Caudate_Body Caudate_Head Caudate_Tail Caudate Anterior_Cingulate Putamen Posterior_Cingulate
 Insula Nucleus_Accumbens Insula Precuneus VS_6mm )
#set ROIs = ( dacc )
#set ROIs = ( MedialFrontalGyrus_01 MiddleTemporalGyrus_01 PostCentralGyrus_01 Putamen_03 SuperiorParietalLobule_01
)
#set ROIs = ( Amygdala_01 CingulateGyrus_01 Claustrum_01 )
#set ROIs = ( NMvstri:Task:Outcome.clust15 )

# CHANGE ME: Specify laterality of the ROIs of interest (left = l, right = r)
#tt daemon
set ROIhemispheres = ( l )

#tt n27 ez ml
#set ROIhemispheres = ( L )
```

- Specify information about your roi extraction output

```
#######################################################
# SPECIFY INFORMATION ABOUT YOUR ROI EXTRACTION OUTPUT
# NOTE: much of this information comes from your individual level analyses
#######################################################


# Specify the sub-brik numbers at the individual level analysis that correspond to the coefficients you used to gene
rate this group-level analysis

#set stim_list = ( 01 04 )
#set stim_list = ( 07 10 13 16 )
set stim_list = ( 13 16 )

#Doors
#LoseGames Ant 01
#WinGames Ant 04
#LoseGames FB Lose 07
#LoseGames FB Win 10
#WinGames FB Lose 13
#WinGames FB Win 16

#Social Doors
#Dislike Ant 01
#Like Ant 04
#Dislike FB Lose 07
#Dislike FB Win 10
#Like FB Lose 13
#Like FB Win 16
```

- Check output: `cd fMRI_ANALYSES/MASKS/ActivationBased/name of this analysis`
  - This is where the text file output from your analysis will be generated
  - If you have more than one ROI, you'll have multiple text files
- Run `Merging_ROI_data.R` to combine if you have more than one ROI
- Convert text file into csv
  - `sed 's/ \+/,/g' <name_of_text_file> > <name of csv>`